

RIBSTONE SOUTH AGCM, AB, CanadaWMO: **710890**Lat: **52.5869N**Lon: **110.3428W**Elev: **644**StdP: **93.82**Time Zone: **-7.00 (NAM)**Period: **08-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-31.6	-28.7	-35.0	0.1	-31.5	-32.1	0.2	-28.7	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.9	29.0	17.6	27.2	16.9	25.5	16.1	19.6	26.1	18.4	24.5	17.3	23.2	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.2	13.3	22.6	16.1	12.3	21.2	15.0	11.5	19.8	58.6	26.1	54.6	24.6	51.1	23.2	24.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) N/A	N/A	N/A	DB	-35.1	32.5	2.0	1.8	-36.6	33.8	-37.7	34.9	-38.9	35.9	-40.3	37.2	(4)
(5)			WB	-35.2	22.1	2.0	1.6	-36.7	23.2	-37.8	24.2	-39.0	25.1	-40.4	26.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	2.8	-11.3	-12.3	-5.1	3.0	11.0	15.0	17.4	16.4	11.6	4.1	-5.3	-12.0	(6)
(7)		DBStd	12.57	8.86	8.04	7.68	5.30	4.24	3.09	2.56	3.14	4.52	5.09	7.16	8.00	(7)
(8)		HDD10.0	3378	661	624	468	215	39	3	0	1	34	193	458	681	(8)
(9)		HDD18.3	5718	919	858	727	461	229	107	48	76	204	442	708	939	(9)
(10)		CDD10.0	747	0	0	0	4	69	152	231	198	83	9	0	0	(10)
(11)		CDD18.3	46	0	0	0	0	1	6	21	16	3	0	0	0	(11)
(12)		CDH23.3	917	0	0	0	3	69	118	310	283	129	4	0	0	(12)
(13)		CDH26.7	203	0	0	0	0	9	17	69	67	42	0	0	0	(13)
(14)	Wind	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(14)
(15)	Precipitation	PrecAvg	385	19	11	15	28	40	70	71	52	29	17	15	15	(15)
(16)		PrecMax	522	39	23	35	81	86	120	137	127	73	41	50	28	(16)
(17)		PrecMin	242	5	4	1	2	6	23	25	6	0	2	3	4	(17)
(18)		PrecStd	77	9	5	8	18	24	27	26	26	22	9	12	6	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.9	6.0	12.9	21.8	27.8	28.7	31.0	31.0	31.2	23.0	12.8	5.7	(19)
(20)			MCWB	3.1	3.0	7.2	9.5	13.7	16.4	18.9	18.0	15.9	12.9	7.3	2.1	(20)
(21)		2%	DB	4.6	3.4	9.8	18.0	25.3	26.6	28.6	28.5	27.1	18.2	8.5	3.7	(21)
(22)			MCWB	2.0	1.2	4.8	8.4	12.8	15.9	19.0	17.7	15.1	10.3	4.8	1.3	(22)
(23)		5%	DB	3.1	1.7	6.7	15.1	22.8	24.6	26.7	26.5	23.9	15.0	5.6	1.4	(23)
(24)			MCWB	1.0	0.0	3.4	6.8	12.0	15.2	18.1	17.1	14.1	8.5	3.0	-0.8	(24)
(25)		10%	DB	1.3	-0.1	4.3	12.3	20.4	22.6	24.8	24.6	20.7	12.2	3.5	-0.9	(25)
(26)			MCWB	-0.4	-1.4	1.7	5.7	10.8	14.3	17.1	16.5	12.7	7.2	1.6	-2.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.4	3.3	7.8	10.5	14.8	18.8	22.2	20.7	17.3	13.6	7.6	2.7	(27)
(28)			MCDWB	6.2	5.9	12.2	20.6	25.0	25.8	28.4	26.6	26.5	22.0	12.1	4.9	(28)
(29)		2%	WB	2.2	1.5	5.0	8.6	13.6	17.3	20.1	19.3	15.7	11.3	5.2	1.2	(29)
(30)			MCDWB	4.4	3.1	8.9	17.2	22.7	23.7	26.7	26.1	25.0	16.3	8.1	3.6	(30)
(31)		5%	WB	1.1	0.1	3.5	7.3	12.7	16.2	18.9	18.0	14.4	9.2	3.2	-0.8	(31)
(32)			MCDWB	2.9	1.7	6.6	14.3	21.2	22.4	24.9	24.1	22.7	14.2	5.2	1.1	(32)
(33)		10%	WB	-0.4	-1.5	1.9	5.9	11.6	15.2	17.8	16.9	13.2	7.5	1.8	-2.4	(33)
(34)			MCDWB	1.3	-0.1	4.2	11.6	19.5	21.2	23.4	22.9	20.1	11.9	3.6	-1.0	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	10.0	10.5	10.3	11.9	14.2	12.4	12.9	13.5	13.9	11.7	8.9	9.1	(35)
(36)			MCDBR	9.7	10.7	12.2	17.2	17.9	16.1	16.7	17.9	20.1	16.6	11.7	10.8	(36)
(37)			MCWBR	7.7	8.9	8.2	8.8	8.0	7.2	8.2	8.3	9.3	9.4	8.1	8.7	(37)
(38)		5% WB	MCDWR	9.9	10.8	11.9	16.1	16.3	13.5	14.9	15.0	18.3	15.6	10.7	10.6	(38)
(39)			MCWBR	8.2	9.1	8.3	8.5	7.8	6.7	8.0	7.6	8.9	9.1	7.7	8.8	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.248	0.289	0.289	0.336	0.359	0.373	0.367	0.379	0.319	0.283	0.258	0.235	(40)	
(41)		taud	2.345	2.257	2.425	2.351	2.333	2.331	2.388	2.324	2.464	2.541	2.457	2.362	(41)	
(42)		Ebn at Noon	772	822	913	900	891	878	878	843	867	841	762	728	(42)	
(43)		Edh at Noon	57	86	89	110	118	119	111	112	86	64	51	47	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.09	2.14	3.61	4.75	5.79	5.97	6.30	5.12	3.62	2.15	1.09	0.81	(44)	
(45)		RadStd	0.10	0.16	0.27	0.45	0.40	0.42	0.32	0.42	0.38	0.30	0.11	0.07	(45)	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
(46)	Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A
(47)	Regional (52 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page